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N+ NETWORK FUNDAMENTALS AND BUILDING NETWORKS DAY:1

TASK1:

Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

RESULT

1. Downloaded and installed Wireshark.
2. Opened Wireshark and selected the active network interface (WiFi/Ethernet).
3. Clicked Start Capture.
4. Opened a web browser and visited instagram.com.
5. Waited for the website to load completely.
6. Returned to Wireshark and clicked Stop Capture.
7. Observed the protocols shown in the packet list PROTOCOL

OBSERVE:

- DNS
- TCP
- UDP
- TLSv1.3
- HTTP/HTTPS
- ARP

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TASKS:2

Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.) Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

RESULT

Protocol	**OSI Layer**	**Reason**
HTTP/HTTPS	Application Layer (Layer 7)	Used for accessing websites and web services.
TCP	Transport Layer (Layer 4)	Provides reliable communication between devices.
IP	Network Layer (Layer 3)	Responsible for routing packets across networks.

TASKS:3

Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

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RESULT

Layer	Daily Life Example
Application	WhatsApp, Instagram
Presentation	Instagram Images, SSL Encryption
Session	Login Session
Transport	TCP Connection
Network	Router and IP Address
Data Link	Wi-Fi Adapter and Switch
Physical	Ethernet Cable, Wi-Fi Signals

TASKS:4

Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words. Think about how your message is typed, packaged, sent over the internet, and received.

RESULT

When I send a message on WhatsApp, it passes through several OSI layers:

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1. Application Layer (Layer 7)

I type a message in WhatsApp and press Send. WhatsApp prepares the message for transmission.

2. Transport Layer (Layer 4)

TCP divides the message into smaller packets and ensures they are delivered correctly.

3. Network Layer (Layer 3)

The IP protocol adds source and destination IP addresses and decides where the packets should go.

4. Data Link Layer (Layer 2)

The Wi-Fi adapter uses MAC addresses to send the packets to the router.

5. Physical Layer (Layer 1)

The data is converted into wireless signals and transmitted through Wi-Fi. The receiver's device receives the signals and reconstructs the message.